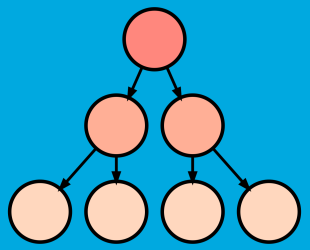
Multilevel Modeling in Clinical Research



Fitting and Interpreting

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Study Overview

The Research Question

> Can clinic-level characteristics predict patient pain?

The Data

25 clinics

492 patients

Patients per clinic: 11-30

DV: Pain Score (0–100 scale)

Clinic-Level Predictors

Predictor	Type
Clinic Size	Continuous
Protocol Type	Binary

Patients in the same clinic may be more similar to each other than to patients in other clinics.

Model Series Overview

Model	Name	Predictors	Parameters	AIC	BIC
Model 0	Single-Level Null	None	2	3389.6	3398.0
Model 1	**Empty MLM (Null)**	**None (+ random intercept)**	**3**	**3283.5**	**3296.1**
Model 2	Main Effects	Clinic Size + Protocol Type	5	3281.3	3302.3
Model 3	Interaction	Size × Protocol Interaction	6	3280.2	3305.4

There are many more possible extensions.

From 1-level to 2-level model

From 1-level to 2-level model

1-level Intercept Only

$$y_i = \beta_0 + \mathrm{e}_i$$

ONE source of variance:

$$\mathrm{e}_i \sim N(0, \sigma_{\mathrm{e}}^2)$$

i = level 1, *j* = level 2

2-level Intercept Only

$$\text{Level 1: } y_{ij} = \beta_{0j} + \mathrm{e}_{ij}$$

$$\text{Level 2: } \beta_{0j} = \gamma_{00} + u_{0j}$$

TWO sources of variance:

$$\mathrm{e}_{ij} \sim N(0, \sigma_{\mathrm{e}}^2)$$

$$u_{0j} \sim N(0, \tau_{00})$$

From 1-level to 2-level model

1-level Intercept Only

R code:

```
m0 <- lm(y ~ 1, data = df)
summary(m0)
```

2-level Intercept Only

R code:

```
m1 <- lmer(y ~ 1 + (1 | cluster), data = df)
summary(m1)
```

Model 0 Single-Level Null

Model 0 Single-Level Null

$$y_i = \beta_0 + \mathrm{e}_i$$

Fixed Effect

Parameter	Estimate	SE
β_0 (Mean)	49.04	0.34

Random Effects

Source	Variance	SD
Total (σ_{e}^2)	57.141	7.559

Empty MLM (Null)

Empty MLM (Null)

Fixed Effect

Parameter	Estimate	SE
γ_{00} (Grand Mean)	49.11	0.86

Random Effects

Source	Variance	SD
Between-Clinic (τ_{00})	16.272	4.034
Within-Clinic (σ_e^2)	41.29	6.426
Total	57.561	

📊 Intraclass Correlation (ICC)

What is the ICC?

$$\mathrm{ICC} = \frac{\tau_{00}}{\tau_{00} + \sigma_{\mathrm{e}}^2} = \frac{16.272}{16.272 + 41.29} = 0.283$$

Three Interpretations

- **Proportion:** 28.3% of variance in pain is **between clinics**
- **Correlation:** Patients from the same clinic correlate at **r = 0.283**
- **Effect Size:** Large clustering effect

> ⚠️ No ICC value is "safe" to ignore.

⚡ Design Effect

Clustering inflates variance (less info than you think)

$$\mathrm{DEFF} = 1 + (\bar{n} - 1) \times \mathrm{ICC}$$

$$= 1 + (19.7 - 1) \times 0.283 = 6.28$$

⚠️ Effective Sample Size

$$N_{\text{eff}} = \frac{N}{\mathrm{DEFF}} = \frac{492}{6.28} \approx 78$$

🤔 Likelihood Ratio Test

Hypothesis:

$H_0 : \tau_{00} = 0$ (no clustering) $\qquad$ $H_1 : \tau_{00} > 0$ (clustering exists)

Result:

$$-2\Delta\mathrm{LL}(df = 1) = 108.13, \mathrm{p} < .001$$

Model improves!

R code:

```
lrt_stat <- -2 * (logLik(Model0) - logLik(Model1))
p_value <- pchisq(lrt_stat, df = 1, lower.tail = FALSE)
```

🤕 Expected Pain Score Range

$$95\%\ \text{CI} = \gamma_{00} \pm 1.96\sqrt{\tau_{00}} \quad = 49.11 \pm 1.96 \times 4.03 = [41.2,\ 57]$$

Expected Distribution of Clinic Means

95% of clinics: 41.2 – 57

Model 2: Main Effects

Adding Level-2 Predictors:

Clinic Size & Protocol Type

Model 2: Equations & Fixed Effects

$$\text{Level 1: } y_{pc} = \beta_{0c} + e_{pc}$$

$$\begin{aligned}\text{Level 2: } \beta_{0c} = \gamma_{00} \\ +\gamma_{01}(\text{size50}_c) \\ +\gamma_{02}(\text{protocol}_c) \\ +U_{0c}\end{aligned}$$

R code:

```
Model2 <- lmer(
  pain_score ~ 1 + size50 +
    protocol_type + (1 | clinic_id),
  data = patient_data, REML = TRUE)
```

Fixed Effects

	Estimate	SE	t	p
γ_{00} Intercept	48.583	1.270	38.261	0.000
γ_{01} Size50	0.920	0.925	0.995	0.331
γ_{02} Protocol	1.078	1.758	0.613	0.546

Intercept: Expected mean pain = 48.58 for clinic size of 50 patients on the Standard protocol

Size50: 0.92 pts per each additional patient past 50

Protocol: Enhanced clinics 1.08 pts higher given clinic size of 50 patients.

Model 2: Variance & Model Fit

LRT: Model 1 vs Model 2

Model	Params	AIC	Chisq	df	p
Model 1 (Empty MLM)	3	3285.02	NA	NA	NA
Model 2 (Main Effects)	5	3287.31	1.713	2	0.4247

Conditional ICC After adding Level-2 predictors:

ICC = 0.258 (25.8% between-clinic variance remains)

Pseudo-R^2 (7.5% of the 28.3% possible)

Component	Base Var	Model 2 Var	Pseudo-R^2
Between-Clinic (L2)	15.535	14.368	0.0751
Within-Clinic (L1)	41.289	41.287	0.0000

Total R^2 (1.92% of the total variance)

Model 3: Interaction

Adding the Size x Protocol Interaction

Model 3: Equations & Fixed Effects

$$\text{Level 2: } \beta_{0c} = \gamma_{00} + \gamma_{01}(\text{size50}_c) + \gamma_{02}(\text{protocol}_c) + \gamma_{03}(\text{size50}_c \times \text{protocol}_c) + U_{0c}$$

Fixed Effects

	Estimate	SE	t	p
γ_{00} Intercept	48.568	1.313	36.988	0.000
γ_{01} Size50	0.834	1.428	0.584	0.565
γ_{02} Protocol	1.090	1.803	0.605	0.552
γ_{03} Size×Protocol	0.153	1.907	0.080	0.937

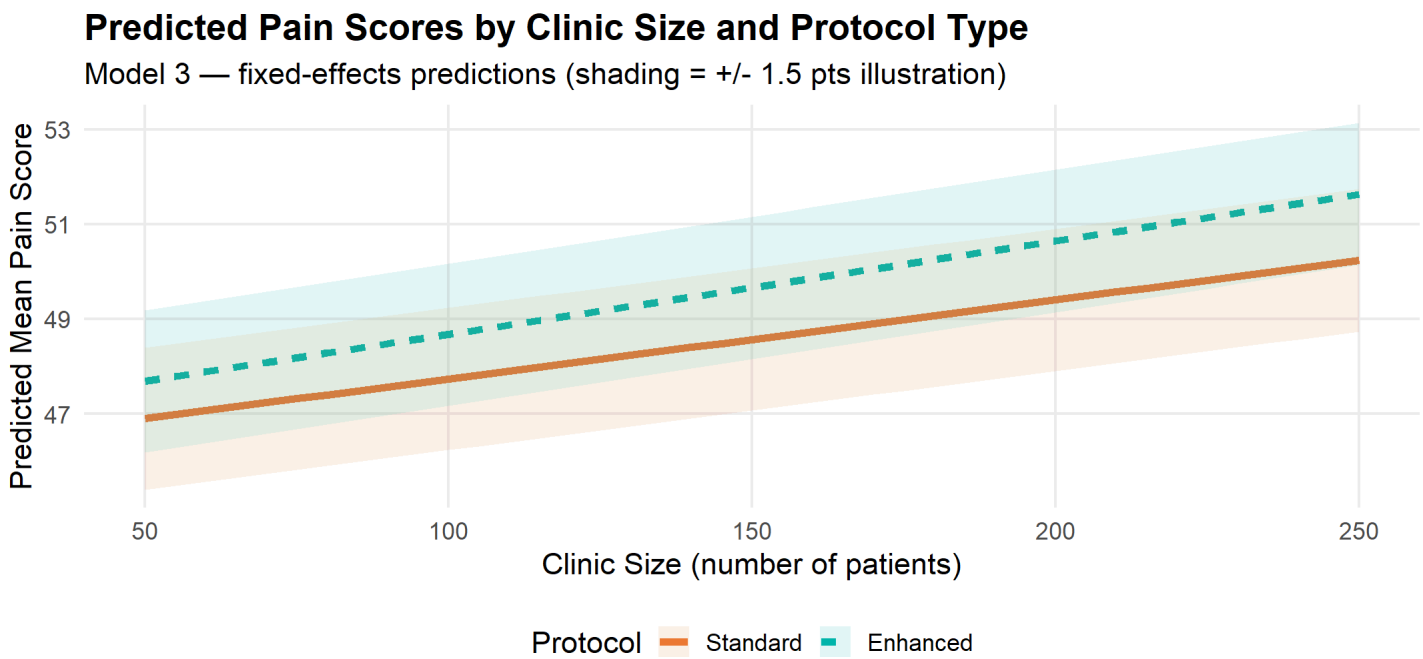
> Interaction γ_{03} = 0.15, p = 0.937

R code:

```
Model3 <- lmer(pain_score ~ 1
+ size50 * protocol_type
+ (1 | clinic_id),
  data = patient_data, REML = FALSE
```

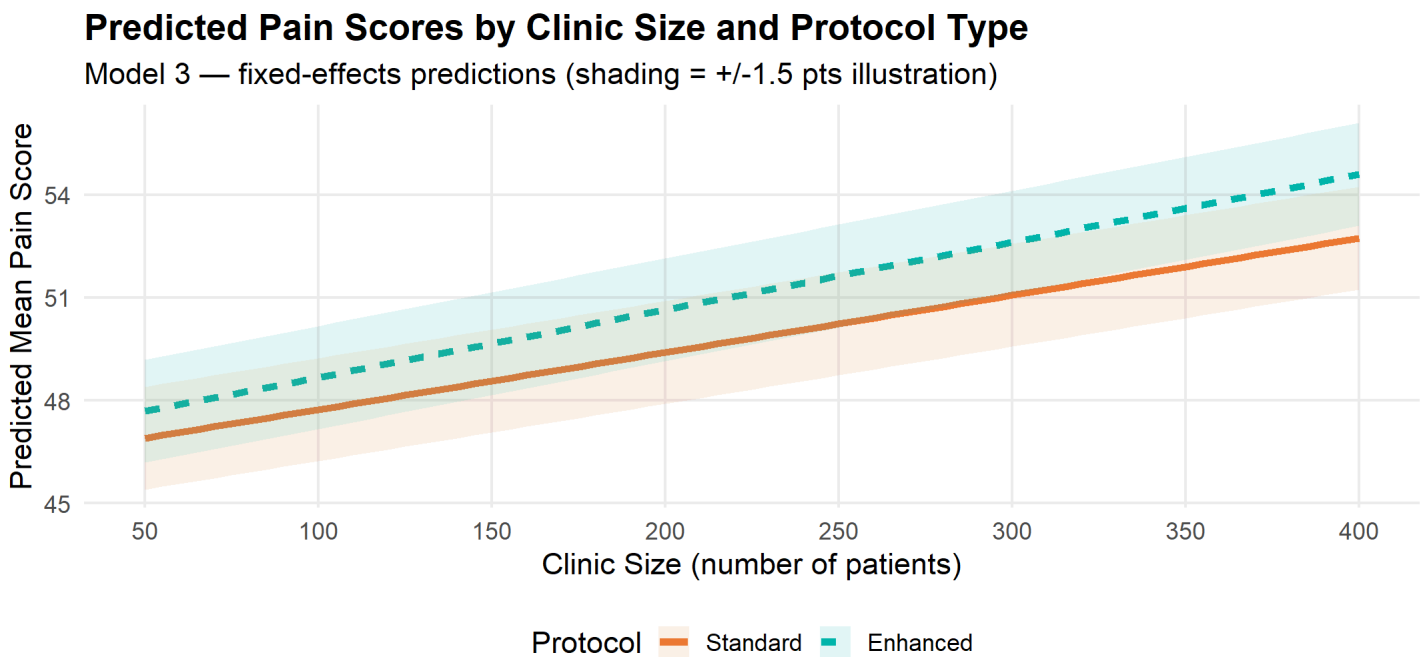
Protocol	Simple Slopes
Standard	γ_{01} = 0.83

Interaction Visualization



> The lines diverge as clinic size increases — this is the interaction. Enhanced protocol shows steeper improvement in larger clinics. Within our observed data range, the lines have not yet crossed, so there is no difference ($p > .05$).

Interaction Visualization



> Extending the x-axis reveals where the lines eventually cross - the point at which one protocol overtakes the other. This is where the Standard vs. Enhanced protocols are significantly different.

Model 3: Fit & Comparison

LRT: Model 2 vs Model 3

Model	Params	AIC	Chisq	df	p
Model 2 (Main Effects)	5	3287.31	NA	NA	NA
Model 3 (Interaction)	6	3289.30	0.009	1	0.9245

Model does not improve.

Pseudo-R2 Comparison

Component	Model 2	Model 3	Δ var
Between-Clinic (L2)	0.0751	0.0759	8e-04
Within-Clinic (L1)	0.0000	0.0000	0e+00

Total R2

Model	Total R^2
Model 2 (Main Effects)	1.92%
Model 3 (Interaction)	1.99%
ΔR^2	0.074

🚀 Next Steps & Extensions

Immediate Next Steps

👤 **Add Level-1 predictors** (patient age, diagnosis severity, visit frequency). This is where 🚀

- 📈 **Test random slopes** - do Level-1 effects vary by clinic?
- 🔀 **Cross-level interactions** - do clinic characteristics moderate patient-level effects?

🆙 **Consider three-level** structure (visits - patients - clinics)

🎓 Thank You!

Additional Questions?

Key Takeaways: 1️⃣ Always check for clustering (ICC) in clinical data 2️⃣ ICC = 0.283 which was huge (still MLM is *almost always* essential anyway) 3️⃣ Effective N = 78, not 492 watch out for the design effect when thinking about powerign your study

The MLM Advantage

✅ Accounts for non-independence ✅ Partitions variance appropriately ✅ Handles unbalanced cluster sizes ✅ Correct standard errors & inference